



Universidade do Minho

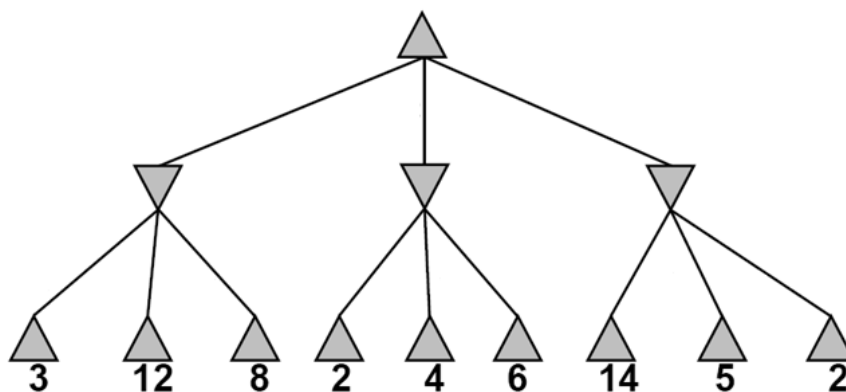
Licenciatura em Engenharia Informática
Mestrado Integrado em Engenharia Informática
Artificial Intelligence

Theoretical-Practical Sheet No. 4

Topic: Search in competitive contexts

Exercise 1

Consider the following game tree.



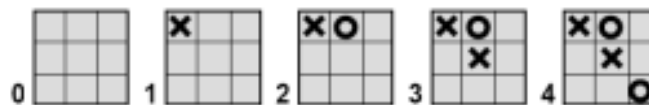
- Apply the minimax algorithm to determine the values of all nodes. Present their values and how they were obtained.;
- Mostre qual o caminho (sequência de estados do jogo) que o “jogador maximizador (MAX)” espera que seja percorrido na árvore acima. Justifique;
- Aplique o algoritmo alfa-beta mostrando, através do desenho da árvore, a evolução dos valores de α e β , os cortes efetuados (justificando) e os valores obtidos para cada nó.



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Exercise 2

Consider the game of tic-tac-toe, in which two players on a 3x3 board take turns placing a piece on an empty square (one marking X, the other O) until they get three pieces in a row, column, or diagonal. The first four moves of a game are illustrated in the following figure.



Considering that the game began with the four moves described in the figure and that the fifth move was placing an X in the top right square (row 1, column 3). Show the complete game tree from the fifth move to a depth of 2. Apply the Minimax algorithm, assuming that the evaluation function is given by the form 100 in case of victory, -100 in case of defeat, 0 in case of a draw, and in the remaining cases by $f(x) = (3O_2 + O_1) - (3X_2 + X_1)$, where X_n is the number of rows, columns and diagonals with a number n of X's and no O's, and O_n is the number of rows, columns and diagonals with a number n of O's and no X's.

What would be the chosen move? And what would be the expected response from player X? In the event of a draw, the preferred moves are those located furthest to the left in the tree.